

Lie Detection in Social VR Using Multimodal Data

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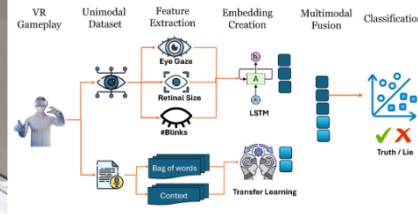
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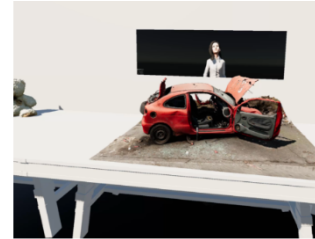
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(a) Side View of Participant During Experiment



(b) Model Pipeline for Lie Detection in VR



(c) VR Game Play for Data Collection

ABSTRACT

Trust is a key element in human relationships and is vital for the success of any interaction. Establishing trust in Social VR is important for promoting collaboration, communication, and social interaction. In our research, we investigate the feasibility of using lie detection techniques in virtual reality to build trust. Our findings suggest that it is possible to detect deception in virtual reality using sensor-detected cues. Using random forest, gaze-based features by themselves could reach an 87% accuracy in detecting lies.

Index Terms: Lie Detection, Social VR

1 INTRODUCTION

Social VR is increasingly engaging users through platforms like VRChat and Horizon World, which facilitate socializing, gaming, education, and business in immersive environments. A major challenge in the immersive environment is establishing trust, a vital component of human interaction, which is hindered by the absence of physical presence and non-verbal cues such as facial expressions and body language [3]. The ability to detect lies in the immersive environment is one mechanism that can be used to increase trust. To the best of our knowledge, there is currently no effective means of detecting lies told by participants in immersive VR environments.

Prior research has examined techniques to detect lies from participants in a non-immersive setting, by measuring changes in psychological behaviors while the participant tells a lie due to a change in cognitive load. Cognitive load theory in lie detection suggests that lying imposes measurable cognitive demands on individuals which can potentially be measured through eye gaze or linguistic features [2].

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This study proposes a novel approach to detect deception in the Social VR using modern VR headset sensors to capture multimodal data, and using a machine learning model trained to detect change in cognitive load while lying. The research is guided by three questions: how users perceive lie detection in Social VR, which features and machine learning models are most effective for detecting lies in VR, and what design considerations must be addressed for implementation, including usability, ethics, and privacy.

To determine the features and machine learning models best fitted for lie detection in VR, we developed a lie detection model and a mechanism to collect training data in the VR environment from 21 users who were recruited to play a VR game where they had to either tell a truth or lie. To learn about how users perceive lie detection in Social VR, we additionally administered a survey to 46 individuals to understand the perception of lie detection.

Our finding showed the feasibility of detecting lies using multimodal data collected from the VR headset with a maximum of 87% accuracy, and determined that while a lie detection mechanism can be used to improve trust in social VR, privacy and ethical issues must be considered.

2 SYSTEM AND MODEL DESIGN

To evaluate a machine learning model in the VR environment, we first collect data in a VR setting, and then construct various models to examine the accuracy of different configurations. To achieve this, we designed and developed a VR game (figure c) in which participants could choose to describe an object accurately or tell a lie, while another player attempted to guess their intention. The game was constructed using the Unity 3D Engine, and rendered on the *Varjo Aero* headset with a resolution of 2880 x 2720 pixels per eye, refresh rate of 90 Hz, and eye-tracking capabilities at 200 Hz with sub-degree accuracy.

Using the data we built machine learning models by using different techniques including: random forest, logistic regression, random forest, nGrams and Naive Bayes, Support Vector machines, with single-modal and multimodal features (Figure b). To enhance the linguistic features, we supplemented data from the Miami University Deception Detection (MUD) Dataset [5].

Study	Our Model				Aboueleine et al [1]				Khan et al [4]			
Best Model	Random Forest				Decision Tree				Random Forest			
Metrics	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
Best Features	0.87	0.87	0.87	0.87	0.80	NA	0.80	NA	0.78	0.72	0.84	0.80
Multimodal Features	0.56	0.56	0.67	0.56	0.68	NA	0.68	NA	0.77	0.70	0.84	0.78

Table 1: Comparison of best-performing models with related works

3 DATA COLLECTION

21 participants (9 male, 12 female) aged between 20 to 50, were recruited to play the VR game that lasted for around 5 minutes. Participants (figure a) spent the 1st minute performing eye calibration on the Varjo's headset, the next 2 minutes receiving instruction from a Non Player Character in the game, and then the next 1 minute describing one of three objects presented in front of them, and in the last 30 seconds answering the NPC whether they lied. Data was collected via the Varjo headset, and recorded using the Varjo Unity SDK, including focus distance, pupil size, gaze direction captured at 5ms intervals. The verbal answers by participants during a 1 minute answer period were also recorded and transcribed. During the game play, 53% of participants told the truth and 47% percent told a lie. The data collected during the game play provides the training data and ground truth used in the model development. Since each participant has approx. 2.5 mins of feed, after cleansing we had 144,160 samples for model training. The experiment was approved by the University's Internal Ethical Review Board (IRB).

4 EVALUATION

To evaluate our model's performance, we withheld 20 percent of the collected data as validation data, and evaluated the model's performance using accuracy, precision, recall, and F-1 score for each model and feature combination. We then evaluated our performance against prior studies in automated lie detection by Aboueleine et al [1] and Khan et al [4] which evaluated eye tracking features, linguistic features, multimodal features.

Given the heterogeneity of the features used, we bench marked our model's performance with the best performing feature and the multimodal model from other studies.

5 RESULTS

Our model with the best features outperformed the best feature and model from referenced studies that leveraged similar techniques to perform lie detection in non-immersive environments as described in Table 1. The uni-modal dataset yielded better accuracy than the multimodal dataset. When we analyzed the uni-modal features using different models, eye gaze data using Random Forest provided the best results at 87% accuracy with 87% recall and precision. Text, analyzed with SVM, yielded the worst result at 33%.

Using pupil size data, Logistic regression model with the Savitzky Golay filter applied achieved 65% accuracy. Random Forest model yielded the highest accuracy at 78%, while the Gradient Boost model achieved 71%. For eye gaze data, the Logistic regression model reached 68% accuracy, but the Random Forest model significantly increased this to 87%.

6 PERCEPTION OF LIE DETECTION IN SOCIAL VR

46 participants (22 male, 22 female, 1 non-binary, and 1 undisclosed), aged between 18 to 54, answered a survey consisting of 14 questions (4 regarding demographics, 4 regarding interactions in Social VR, 2 questions regarding whether they have lied or been lied to, and 4 questions regarding the usefulness of a lie detection feature in VR). 85% of the participants had prior VR experience.

When asked how useful respondents would find a feature which would determine if an avatar is lying, 65% respondents were sure they would find it useful while another 20% found it difficult to establish whether it is useful or not. Only 14% said that they would not find it useful. Hence, we know that such a feature would be useful to people in the future and has great potential.

Respondents expressed significant concerns regarding the potential misuse of lie detection features. They agreed that bad actors could exploit others in Social VR settings for scams and acknowledged the risk of falling victim to social engineering. The average concern rating for the negative consequences of false lie detection was 7.35 on a scale of 1 to 10, suggesting a high concern. Participants were particularly worried about being incorrectly identified as liars and noted that the system's effectiveness might vary across different cultures and languages.

7 DISCUSSION AND FUTURE WORK

Our model results demonstrate the feasibility of using analytical models to detect deception in the Social VR environments, particularly through eye gaze analysis using a random forest model, which demonstrated a relatively high accuracy.

Our survey indicates that users value the ability to detect lies in social VR where establishing trust is crucial. Based on user feedback, we propose design considerations which includes: allowing the user to turn the functionality on and off on demand, build models for different languages, and take measurements to minimize false positives.

Future research could focus on incorporating additional features to enhance model accuracy, leveraging large language models (LLMs) to improve lie detection precision, or exploring system architectures that enable real-time lie detection.

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